

Q.1 If $Ax = B$ a linear system such that

$A^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, Then $x = ??$

$$A^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$x = A^{-1} \cdot B$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \rightarrow x = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

Q. 2 Use one iteration bisection method to estimate the solution for $x - 2^x = 0$ for $0 \leq x \leq 5$

$$f(0) = -1 \quad \text{et} \quad f(1) = 0.5 \quad \rightarrow \quad 0.5 * -1 = -0.5 < 0 \quad \checkmark$$

1st iteration

$$x_0 = \frac{0+1}{2} = 0.5$$



Q. 3] If the true value of x is 6 and the estimated value of it is 5.5 the percent relative true error is??

Sol. $\left| \frac{6 - 5.5}{6} \right| \times 100\% = 8.3\%$

Q. 4] If the fourth and fifth approximations of the length of a bridge are: 8 cm and 10 cm, respectively then the relative approximate error is equal to???

Sol. $\left| \frac{10 - 8}{10} \right| \times 100\% = 20\%$

Q. 5] Determinantes $\begin{vmatrix} a & 0 & 0 \\ 2 & a & 0 \\ -1 & 2 & a \end{vmatrix} = ???$

Sol. $\begin{vmatrix} a & 0 & 0 \\ 2 & a & 0 \\ -1 & 2 & a \end{vmatrix}$

$+ a [a^2 - 0] - 0 [2a - 0] + 0 [4 - a]$
 $= a^3$

* Determinates $\Rightarrow a^3$



Q.6 Apply Newton-Raphson method on $x^5 - 10 = 0$
 Where the initial guess $x_0 = 1$ in order to find x_3 ???
 [Round off your answer to 4 decimal places] ??

Sol. $f(x) = x^5 - 10$ & $f'(x) = 5x^4$

* $f(x_0) = f(1) = -9$

$f'(x_0) = f'(1) = 5 \rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$x_1 = 1 - \left(\frac{-9}{5}\right) \rightarrow x_1 = 2.8$

$f(x_1) = f(2.8) = 162.1037$

$f'(x_1) = f'(2.8) = 307.328 \rightarrow x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$x_2 = 2.2725$ & $f(x_2) = 50.6117$

$f'(x_2) = 133.3568$

$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$

$\rightarrow x_3 = 1.893$



Q. 7 If the fourth and fifth approximations of the length of the bridge are: 49 m and 50 m, respectively then the relative approximate error is equal to:

Sol. $\left| \frac{50 - 49}{50} \right| \times 100\% \rightarrow \boxed{= 2\%}$

Q. 8 Apply Newton-Raphson method on $x^5 - 10 = 0$.
Where the initial guess $x_0 = 3$ in order to find x_3 ???
[Round off your answer to 4 decimal places].

Sol. $f(x) = x^5 - 10$

$f'(x) = 5x^4$

$f(x_0) = \cancel{f(3)} f(3) = \boxed{233}$

$f'(x_0) = f'(3) = \boxed{405}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3 - \frac{233}{405} \quad \boxed{x_1 = 2.4247}$

$f(x_1) = 73.8079$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \rightarrow \boxed{x_2 = 1.9976}$

$f'(x_1) = 172.8207$

$f(x_2) = 21.8098$

$f'(x_2) = 79.6143$

$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \rightarrow \boxed{x_3 = 1.7237}$



Q.9 Solving the equation, $f(x)=0$, by newton-Raphson method if $x_0=1$, $f'(x_0)=2$, $x_1=3$ find $f(x_0)=???$

Sol. $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \rightarrow 3 = 1 - \frac{f(x_0)}{2}$

$f(x_0) = -4$

Q.10 Use first iteration newton Raphson method to estimate a root of the Polynomial $f(x)=x^2$ start with the initial estimate $x_0=2$??

$f(x)=x^2$, $f'(x)=2x$

Sol. $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \rightarrow x_1 = 2 - \frac{4}{4} \rightarrow x_1 = 1$



Q. 15 If $AX=B$ is a system of linear equations such that A is 4×5 -matrix and X is 5×1 -matrix then this system has:

b.

- a. 4 equations with 4 unknowns. b. 4 equations with 5 unknowns.
c. 5 equations with 5 unknowns. d. 5 equations with 4 unknowns.

Q. 16 If $AX=B$ is a system of linear equations such that A is 6×2 -matrix and X is 2×1 -matrix then this system has:

c.

- a. 2 equations with 2 unknowns. b. 2 equations with 6 unknowns.
c. 6 equations with 2 unknowns. d. 6 equations with 6 unknowns.

Q. 17 $[A^T$: The transpose of the matrix $A]$

A matrix A is called symmetric if:

d.

a. $A^T = -A$

b. $A^T = A^{-1}$

c. $AA^T = I$

d. $A^T = A$

Q. 18 The system: $x+y=4$ & $2x+2y=8$ select one:

a. has no solution. b. has unique solution. d.

c. is not linear system. d. has infinite solutions.



رقم الصفحة (٧)

SPU